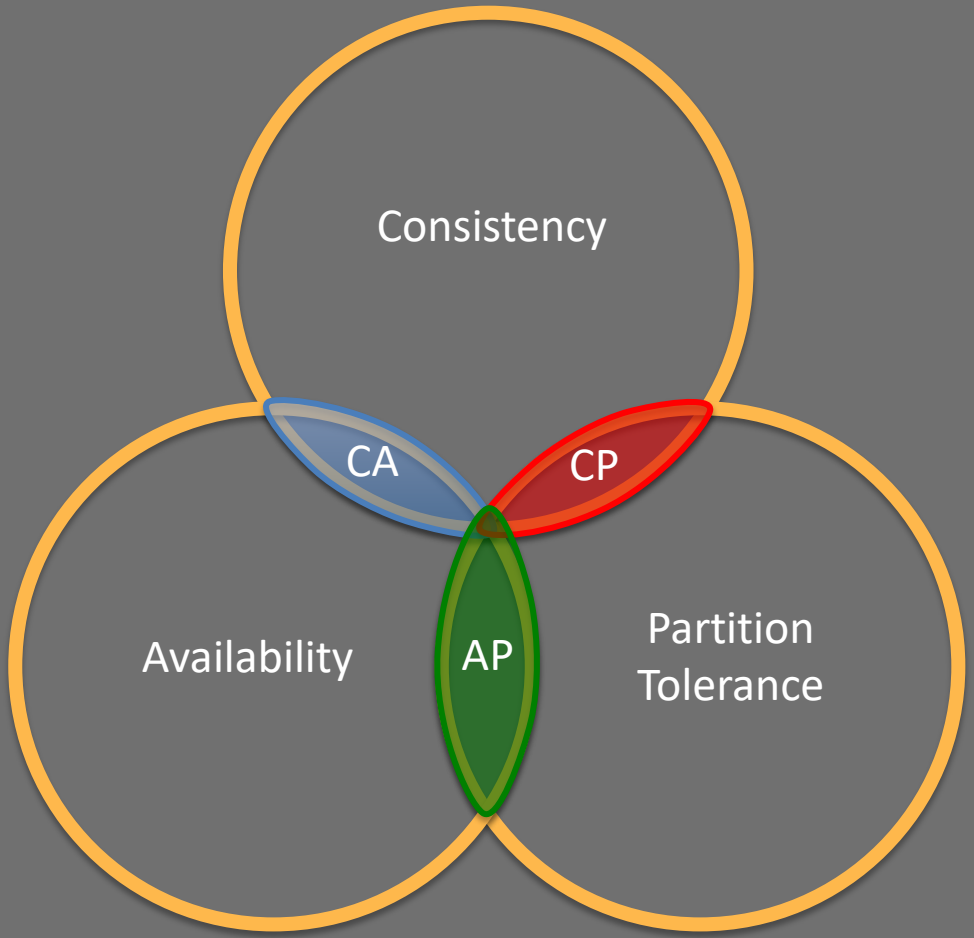
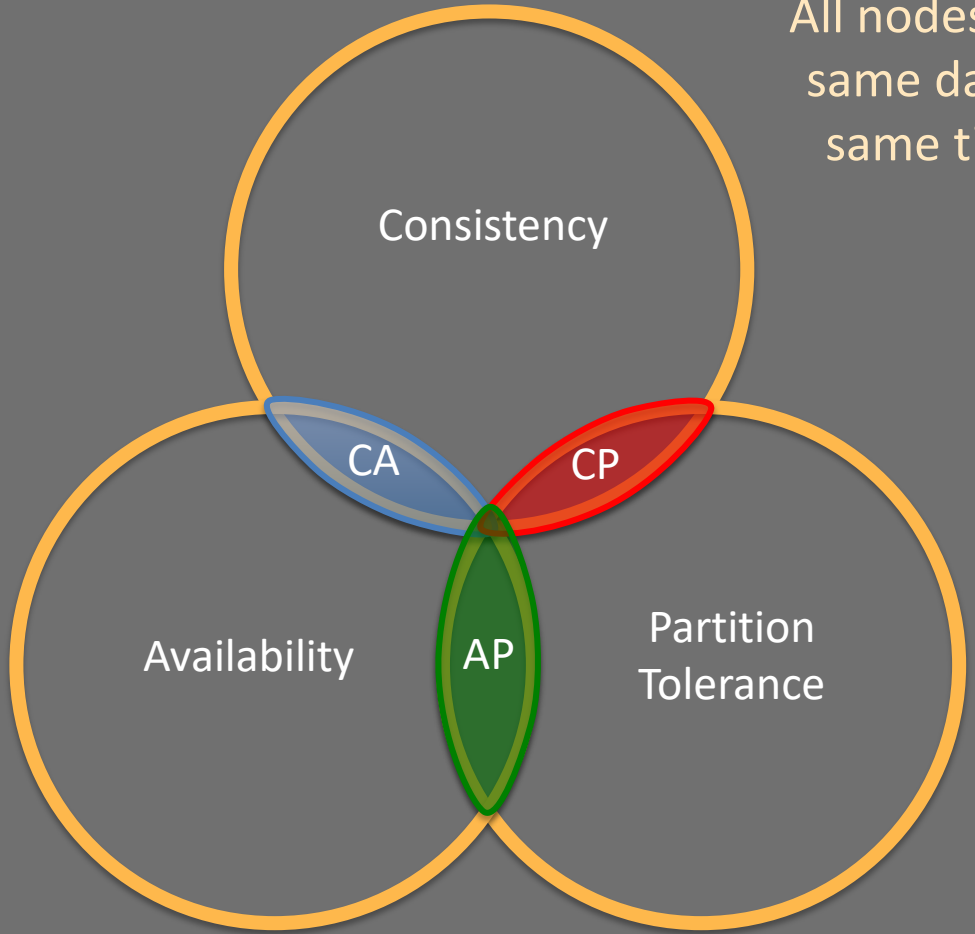
CAP Theorem



Can’t have all three properties.

All nodes have same data at same time.

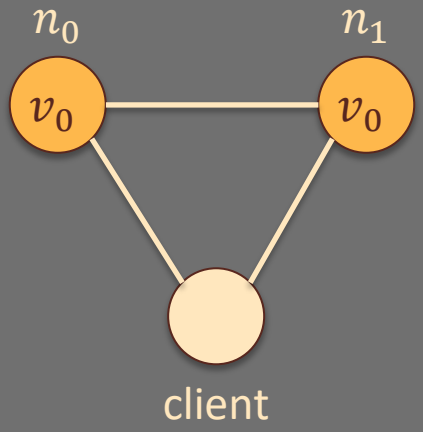


All requests handled.

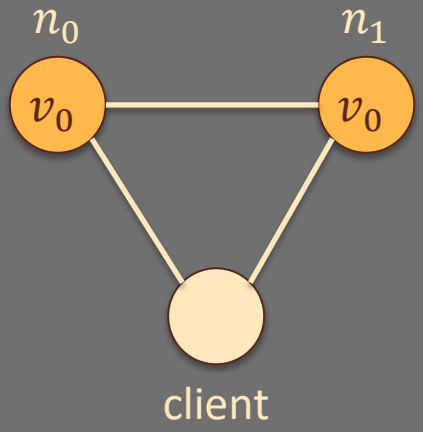
Entire system doesn't fail if some nodes fail.

WHY CAN’T WE HAVE IT ALL?

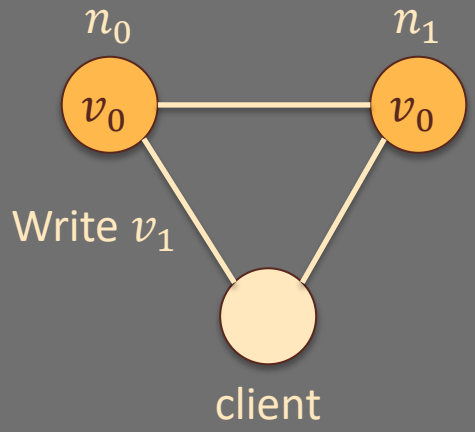
A simple distributed model



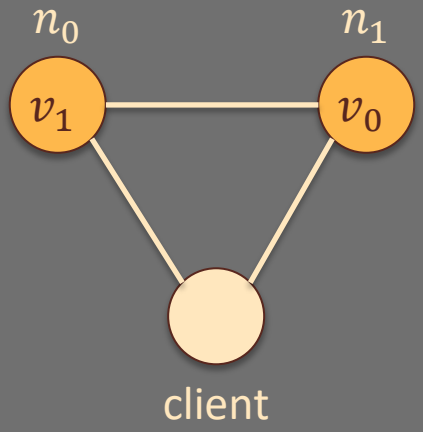
Example of inconsistency



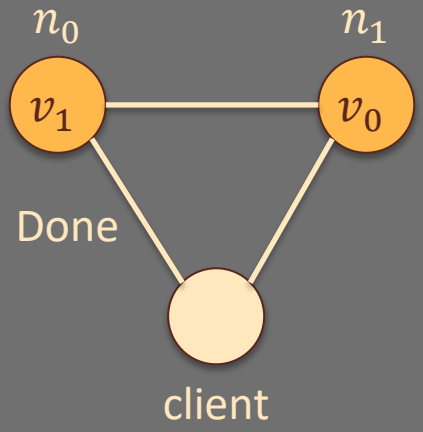
Example of inconsistency: 1



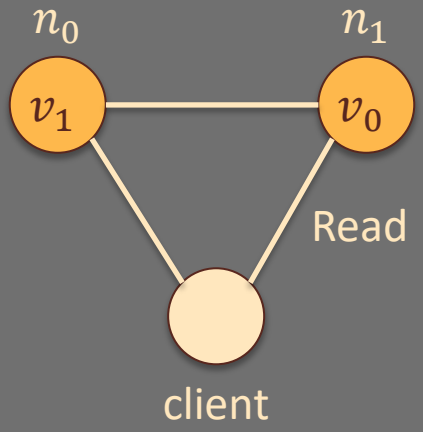
Example of inconsistency: 2



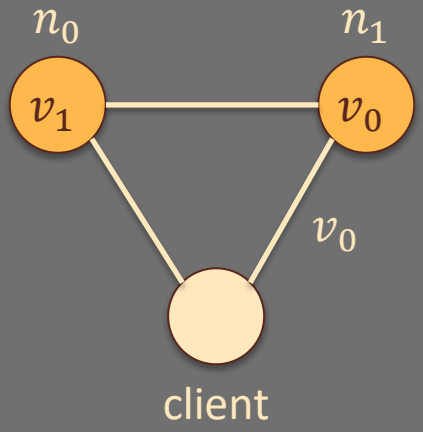
Example of inconsistency: 3



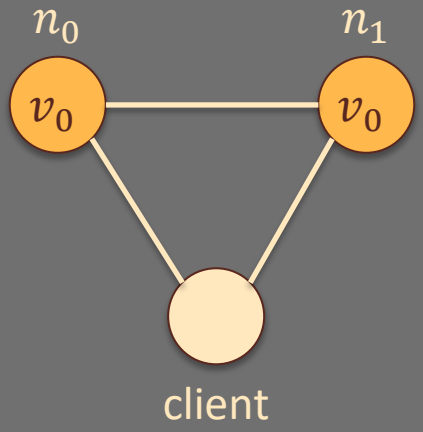
Example of inconsistency: 4



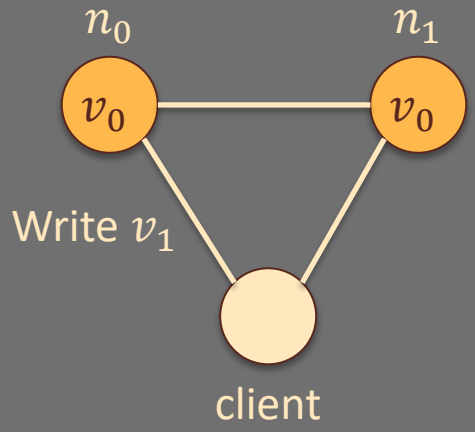
Example of inconsistency: 5



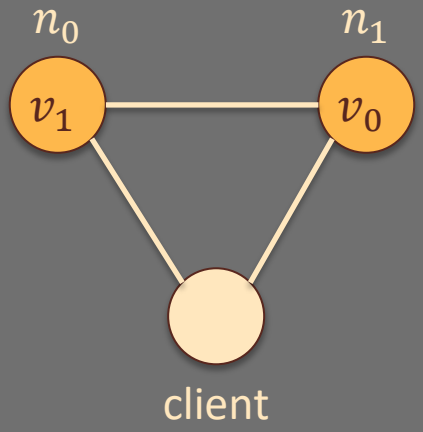
Example of consistency



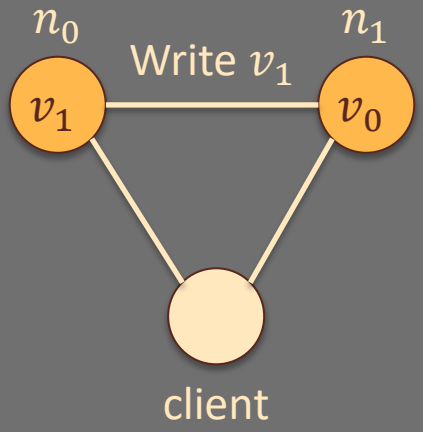
Example of consistency: 1



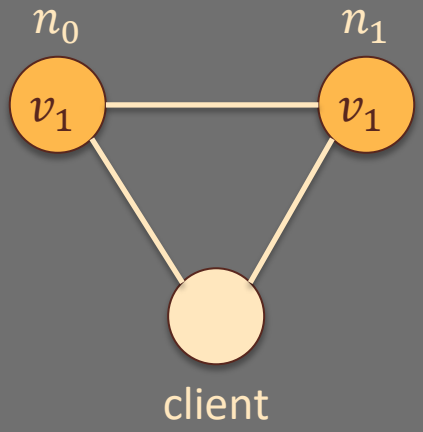
Example of consistency: 2



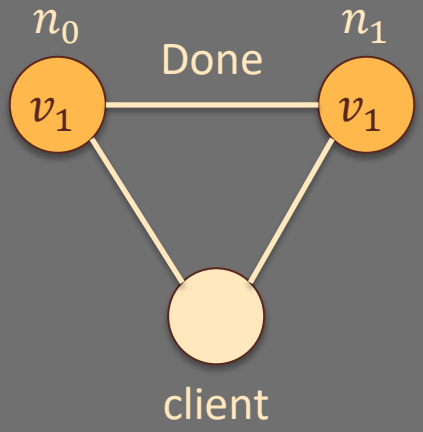
Example of consistency: 3



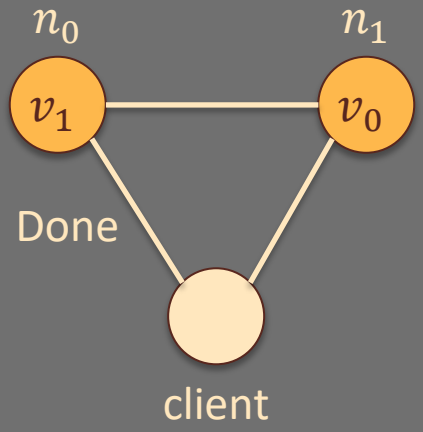
Example of consistency: 4



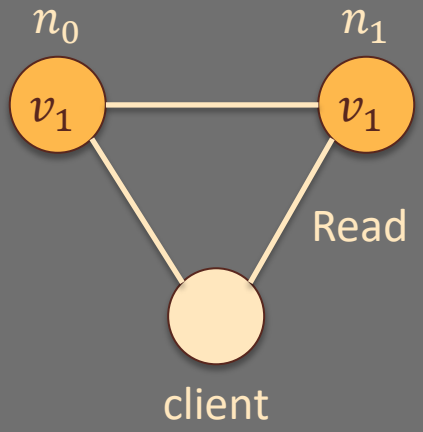
Example of consistency: 5



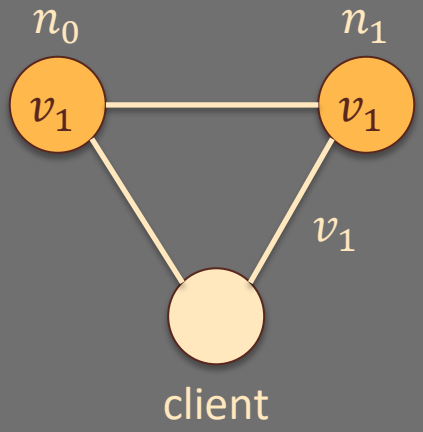
Example of consistency: 6



Example of consistency: 7



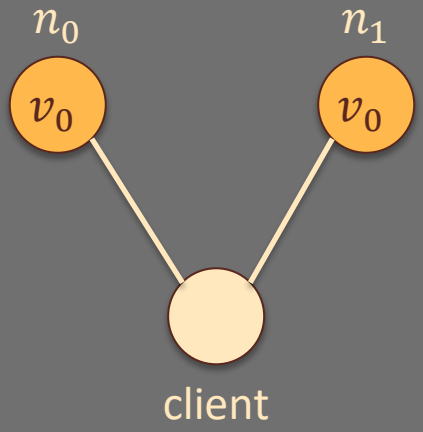
Example of consistency: 8



Availability

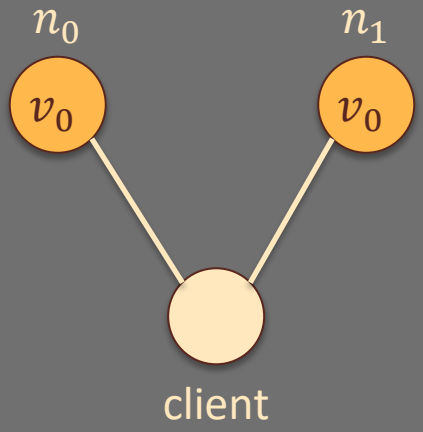
Every request gets a response.

Example of partition tolerance



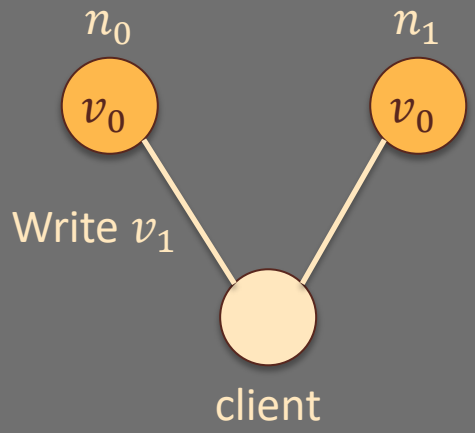
Operations should still work.

Can’t satisfy all three

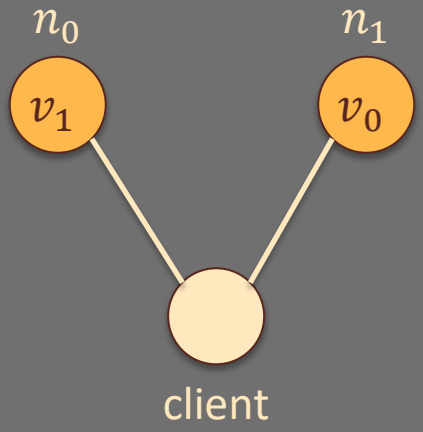


Assume we can. See that this fails.

Can’t satisfy all three: 1

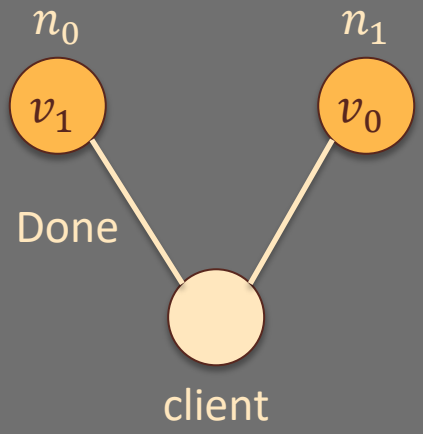


Can’t satisfy all three: 2

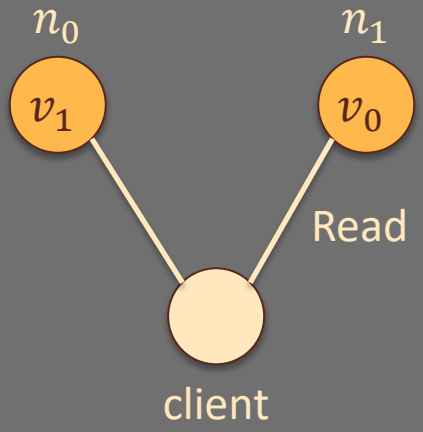


Can’t satisfy all three: 3

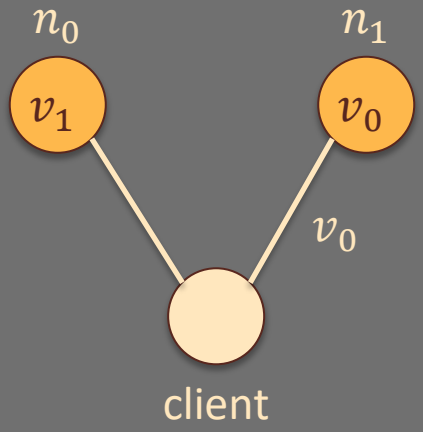
Availability: Must respond.
Partition: Can’t update n_1.



Can’t satisfy all three : 4



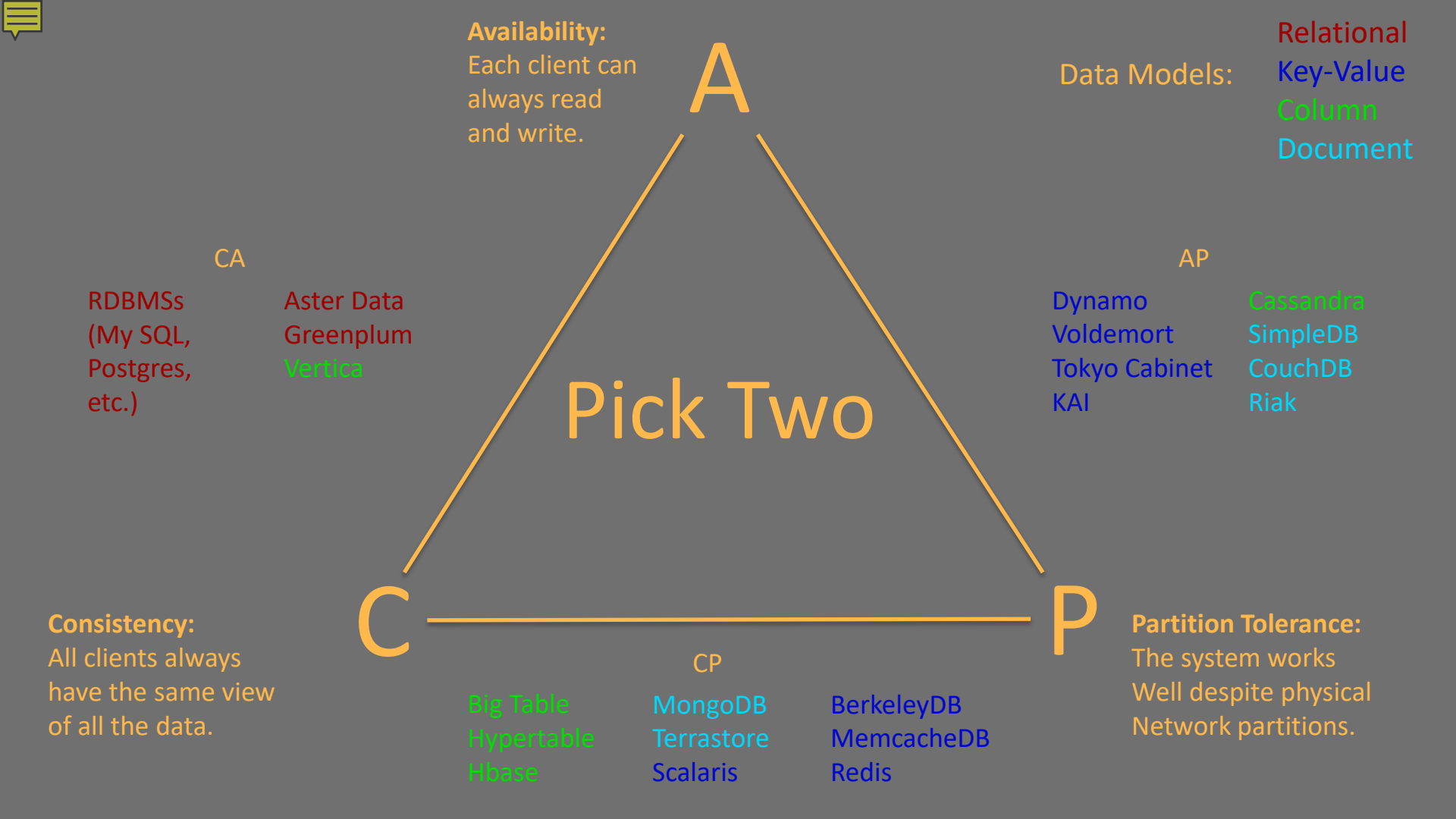
Can’t satisfy all three : 5



Availability: Must respond.
Partition: Can’t update n_1.

Examples of choosing tradeoffs

- Web photo sharing
  - Want fast response
  - Displayed photo can be stale
- Airline reservation
  - Near empty? Double-booking less likely. Emphasize speed.
  - Near full? Emphasize consistency.
- Online retail
  - Shopping cart must be available
  - Product info can be stale
  - Checkout must be consistent


Availability:
Each client can always read and write.
A
Data Models:
Relational
Key-Value
Column
Document
CA
RDBMSs (My SQL, Postgres, etc.)
Aster Data
Greenplum
Vertica
Pick Two
AP
Dynamo
Voldemort
Tokyo Cabinet
KAI
Cassandra
SimpleDB
CouchDB
Riak
Consistency:
All clients always have the same view of all the data.
C
P
Partition Tolerance:
The system works Well despite physical Network partitions.
CP
Big Table
Hypertable
Hbase
MongoDB
Terrastore
Scalaris
BerkeleyDB
MemcacheDB
Redis